In 1905, **Blackman** put forth the importance of smallest factor. **Blackman's law of limiting factor** is actually a modified Law proposed by **Liebig's Law of minimum**. According to Blackman, *"When a process is conditioned as to its rapidity by a number of separate factors, the rate of the process is limited by the pace of the lowest factor"*. To conclude in an easy way *"at any given point of time the lowest factor among essentials will limit the rate of photosynthesis"*. For example, when even sufficient light intensity is available, photosynthesis may be low due to low CO_2 in the atmosphere. Here, CO_2 acts as a limiting factor. If CO_2 is increased in the atmosphere the rate of photosynthesis also increases. Further increase in photosynthesis is possible only if the available light intensity is also increased proportionately (Figure 13.21).

Factors affecting photosynthesis are further grouped into External or Environmental factors and Internal factors.

I. **External factors:** Light, carbon dioxide, temperature, water, mineral and pollutants.
II. **Internal factors:** Pigments, protoplasmic factor, accumulation of carbohydrates, anatomy of leaf and hormones.

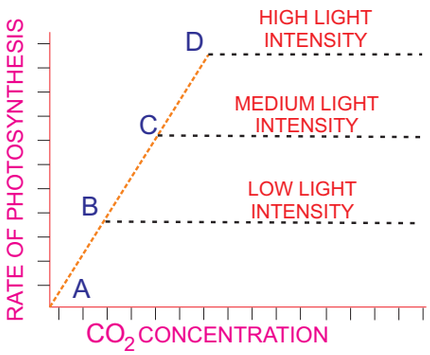


Figure 13.21: Blackman's Law of Limiting Factors

13.15.1. External factors

1. Light

Energy for photosynthesis comes only from light. Photooxidation of water and excitation of pigment molecules are directly controlled by light. Stomatal movement leading to diffusion of CO_2 is indirectly controlled by light.

a. Intensity of Light:

Intensity of light plays a direct role in the rate of photosynthesis. Under low intensity the photosynthetic rate is low and at higher intensity photosynthetic rate is higher. It also depends on the nature of plants. Heliophytes (Bean Plant) require higher intensity than Sciophytes (*Oxalis*).

b. Quantity of Light:

In plants which are exposed to light for longer duration (Long day Plants) photosynthetic rate is higher.

c. Quality of light:

Different wavelengths of light affect the rate of photosynthesis because pigment system does not absorb all the rays equally. Photosynthetic rate is maximum in blue and red light. **Photosynthetically Active Radiation** (PAR) is between 400 to 700 nm. Red light induces highest rate of photosynthesis and green light induces lowest rate of photosynthesis.

2. Carbon dioxide

CO_2 is found only 0.3 % in the atmosphere but plays a vital role. Increase in concentration of CO_2 increases the rate of photosynthesis (CO_2 concentration in the atmosphere is 330 ppm). If concentration is increased beyond 500ppm, rate of photosynthesis will be affected showing the inhibitory effect.

3. Oxygen

The rate of photosynthesis decreases when there is an increase of oxygen concentration. This Inhibitory effect of oxygen was first discovered by **Warburg** (1920) using green algae *Chlorella*.

4. Temperature

The optimum temperature for photosynthesis varies from plant to plant. Temperature is not uniform in all places. In general, the optimum temperature for photosynthesis is 25°C to 35°C. This is not applicable for all plants. The ideal temperature for plants like *Opuntia* is 55°C, Lichens 20°C and Algae growing in hot spring photosynthesis is 75°C. Whether high temperature or low temperature it will close the stomata as well as inactivate the enzymes responsible for photosynthesis (Figure 13.22).

XIth_BIO-BOTANY_Ch_13-EM.indd 257 5/17/2020 11:28:57 PM